Tanmay Singh

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Data Engineer with experience designing and scaling ETL pipelines, relational schemas, and cloud data infrastructure for AI-driven products and enterprise financial systems. Skilled in building reliable, high-throughput data platforms across Airflow, dbt, BigQuery, PostgreSQL, Supabase, and AWS, with a track record of cutting pipeline runtime by 40% and processing 10M+ records monthly at production scale.

EDUCATION

Arizona State University — Master of Science in Data Science, Analytics and Engineering
Manipal University, Jaipur — Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

Vault IQ **Dec 2025 – May 2026**
Data Engineer (AI and Backend Systems) *Remote*

- Designed and implemented relational schemas and storage for user, brand, product, and social data in Supabase (PostgreSQL), giving 4+ product modules one structured data flow
- Worked full stack across the data layer, backend services, and product UI of an AI-powered brand development platform integrating OpenAI, Gemini, and Flux Pro
- Sped up the brand-generation pipeline that orchestrates multi-model LLM and image-generation calls for content and product visualization
- Built a photoshoot agent that stages generated product images in marketing-ready settings such as hero shots, using reasoning-based prompt orchestration while preserving product fidelity
- Built and deployed a production chatbot with a custom knowledge-base integration layer for contextual querying and FAQ automation
- Built the membership dashboard backend serving brand portfolio, product catalog, and revenue metrics in real time, integrating 6+ external services and APIs

Capgemini Pvt Ltd **Oct 2020 – Mar 2023**
Data Engineer – HSBC UK (Application Data Management) *United Kingdom*

- Designed and orchestrated scalable ETL pipelines in Apache Airflow (DAG-based workflows) for batch processing of HSBC's customer credit data systems
- Managed and processed 10M+ customer records monthly across Google BigQuery and PostgreSQL environments, ensuring data consistency, integrity, and availability
- Developed transformation and validation workflows in Python and PySpark, with data quality checks aligned to business-critical credit risk models
- Optimized BigQuery queries and storage structures and implemented SCD logic, reducing batch pipeline runtime by 40% and meeting ETL SLAs
- Built monitoring into pipeline runs to catch data anomalies pre-emptively, improving the reliability of production credit data systems

PROJECTS

Mettle | Gamified Workout Tracker (in development)

- A workout tracker that turns training into a game: workouts earn XP, and hitting real benchmarks ranks you from E to S
- Offline-first: workouts log to on-device SQLite and sync to Supabase Postgres, secured with row-level security
- A shared TypeScript engine recomputes XP server-side in an Edge Function, so leaderboards can't be gamed from the phone

CheckYourAQI | Air Quality Forecasting Platform (live)

- Forecasts air quality (PM2.5) 1–3 days ahead for 70 Delhi NCR monitoring stations, on a public dashboard and API
- Hourly Airflow ingestion of air-quality, weather, and fire data into an S3/Iceberg lakehouse with 27 dbt models and 44 tests
- Weekly model retraining on GitHub Actions; runs unattended on AWS for ~$21/month with failure alerts

Real-Time Clickstream Analytics Pipeline on AWS

- Simulates website click events and processes them in near real time into a queryable analytics dataset on AWS
- Kinesis and Lambda land raw events in S3; AWS Glue ETL cleans and partitions them by date for Athena
- Step Functions orchestrates the run end to end, with SNS alerts on success or failure

CheckYourData | AI-Suggested Data Quality Checks

- Upload a dataset and an AI agent suggests data-quality checks for it, which you review, run, and track over time
- FastAPI and Postgres backend stores check runs and tracks drift; one Docker service serves the API and React app
- Suggestions are cached by a schema hash, cutting repeat calls from ~4 s to ~40 ms

SKILLS

Languages: Python, SQL, PySpark, TypeScript
Data Platforms & Warehousing: PostgreSQL, Google BigQuery, Supabase, Amazon S3, Apache Iceberg, Athena, Data Modeling, Schema Design
Pipelines & Orchestration: Apache Airflow (DAG-based ETL), dbt, AWS Glue, Kinesis, Step Functions, Batch & Streaming, Data Quality Checks
Cloud & Tools: AWS (EC2, Lambda), Google Cloud, Docker, GitHub Actions, Git, FastAPI, REST APIs, LLM/Generative AI Integration